

lab07_hannah_miranda_nakamura

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Laboratório 07

SQLite - Parte II

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Atividade

```
library(RSQLite)
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.6.1

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
path <- "datasets" # caminho até o arquivo
fname <- file.path(path, "uwmadison.sqlite3") # nome do arquivo com o seu caminho
conn <- dbConnect(SQLite(), fname) # variável conn armazena a conexão
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

Quem são os professores

```
SELECT DISTINCT(instructors.name) FROM instructors
INNER JOIN teachings ON teachings.instructor_id = instructors.id
INNER JOIN sections ON sections.uuid = teachings.section_uuid
INNER JOIN subject_memberships ON subject_memberships.course_offering_uuid = sections.course_offering_uuid
INNER JOIN subjects ON subjects.code = subject_memberships.subject_code
WHERE subjects.abbreviation = 'STAT'
```

Table 1: Displaying records 1 - 10

name
MINJING TAO
DONALD PORTER
SHENG WANG
KUNLING HUANG
DONG XIA
JENNIFER NGUYEN
SEULKEE YUN
DUZHE WANG
YONGFENG WU
JARED BROWN

Note que a tabela já indica a presença de 377 linhas totais, portanto, há 377 professores. Mesmo assim, optei por criar um código apenas para essa parte com o intuito de treino.

```
SELECT COUNT(DISTINCT(instructors.name)) FROM instructors
INNER JOIN teachings ON teachings.instructor_id = instructors.id
INNER JOIN sections ON sections.uuid = teachings.section_uuid
INNER JOIN subject_memberships ON subject_memberships.course_offering_uuid = sections.course_offering_uuid
INNER JOIN subjects ON subjects.code = subject_memberships.subject_code
WHERE subjects.abbreviation = 'STAT'
```

Table 2: 1 records

COUNT(DISTINCT(instructors.name))
377

- O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

- Quem é o professor mais difícil?

```
SELECT instructors.name, SUM(
a_count * 4.0 + ab_count * 3.5 + b_count * 3.0 + bc_count * 2.5 + c_count * 2.0 + d_count * 1.0 + f_count * 0.0) / SUM(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) AS MEDIA_NOTAS
INNER JOIN teachings ON teachings.instructor_id = instructors.id
```

```

INNER JOIN sections ON sections.uuid = teachings.section_uuid
INNER JOIN course_offerings ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN grade_distributions ON grade_distributions.course_offering_uuid = course_offerings
INNER JOIN subject_memberships ON subject_memberships.course_offering_uuid = sections.course
INNER JOIN subjects ON subjects.code = subject_memberships.subject_code
WHERE subjects.abbreviation = 'STAT'
GROUP BY instructors.name
HAVING SUM(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0
ORDER BY MEDIA_NOTAS
LIMIT 1

```

Table 3: 1 records

name	MEDIA_NOTAS
JAMES D KUELBS	2.607692

Note que JAMES D KUELBS é o mais difícil

- Quem é o professor mais fácil?

```

SELECT instructors.name, SUM(
a_count * 4.0 + ab_count * 3.5 + b_count * 3.0 + bc_count * 2.5 + c_count * 2.0 + d_count * 1
)/ SUM(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) AS MEDIA_NOTAS
INNER JOIN teachings ON teachings.instructor_id = instructors.id
INNER JOIN sections ON sections.uuid = teachings.section_uuid
INNER JOIN course_offerings ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN grade_distributions ON grade_distributions.course_offering_uuid = course_offerings
INNER JOIN subject_memberships ON subject_memberships.course_offering_uuid = sections.course
INNER JOIN subjects ON subjects.code = subject_memberships.subject_code
WHERE subjects.abbreviation = 'STAT'
GROUP BY instructors.name
HAVING SUM(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0
ORDER BY MEDIA_NOTAS DESC
LIMIT 5

```

Table 4: 5 records

name	MEDIA_NOTAS
YAJUAN SI	4.000000
SRIKANTHMADHAVAN ARAVAMUTHAN	4.000000

name	MEDIA_NOTAS
GUANHUA CHEN	4.000000
GRACE WAHBA	3.971154
YU MENGANG	3.935780

Notes que YAJUAN SI, SRIKANTHMADHAVAN ARAVAMUT e GUANHUA CHEN são igualmente mais fáceis

- Qual é a disciplina mais difícil?

```
SELECT courses.name, SUM(
a_count * 4.0 + ab_count * 3.5 + b_count * 3.0 + bc_count * 2.5 + c_count * 2.0 + d_count * 1.0
)/ SUM(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) AS MEDIA_NOTAS
FROM courses
INNER JOIN course_offerings ON course_offerings.course_uuid = courses.uuid
INNER JOIN grade_distributions ON grade_distributions.course_offering_uuid = course_offerings.uuid
INNER JOIN subject_memberships ON subject_memberships.course_offering_uuid = course_offerings.uuid
INNER JOIN subjects ON subjects.code = subject_memberships.subject_code
WHERE subjects.abbreviation = 'STAT'
GROUP BY courses.uuid, courses.name
HAVING SUM(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0
ORDER BY MEDIA_NOTAS
LIMIT 1
```

Table 5: 1 records

name	MEDIA_NOTAS
Introduction to the Theory of Probability	2.886374

- Qual é a disciplina mais fácil?

```
SELECT courses.name, SUM(
a_count * 4.0 + ab_count * 3.5 + b_count * 3.0 + bc_count * 2.5 + c_count * 2.0 + d_count * 1.0
)/ SUM(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) AS MEDIA_NOTAS
FROM courses
INNER JOIN course_offerings ON course_offerings.course_uuid = courses.uuid
INNER JOIN grade_distributions ON grade_distributions.course_offering_uuid = course_offerings.uuid
INNER JOIN subject_memberships ON subject_memberships.course_offering_uuid = course_offerings.uuid
INNER JOIN subjects ON subjects.code = subject_memberships.subject_code
WHERE subjects.abbreviation = 'STAT'
```

```
GROUP BY courses.uuid, courses.name
HAVING SUM(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0
ORDER BY MEDIA_NOTAS DESC
LIMIT 5
```

Table 6: 5 records

name	MEDIA_NOTAS
Nonparametric Statistics and Machine Learning Methods	4.000000
Sample Survey Theory and Method	4.000000
Data Science Practicum	4.000000
Empir Proc&Semiparmtrc Infernc	4.000000
Experimental Design I	3.984127

Note que Nonparametric Statistics and Machine Learning Methods, Sample Survey Theory and Method, Data Science Practicum e Empir Proc&Semiparmtrc Infernc são igualmente as mais fáceis.

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-22 01:03:36 -03"
```